

Wave equation: analysis and discretization

Introduction to Numerical Relativity

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First order reduction of the wave equation (in 1+1) I

The wave field ψ only depends on t and one spatial coordinate, say x :

$$\psi(t, x)$$

The wave equation reduces to ($\cdot$ for time derivatives and $'$ for spatial ones):

$$\frac{\partial^2 \psi}{\partial t^2} = \frac{\partial^2 \psi}{\partial x^2} \quad \leftrightarrow \quad \ddot{\psi} = \psi''$$

To apply “textbook” methods, it is convenient to write it in first order form. Introduce:

$$\begin{aligned}\pi &= \dot{\psi} \\ \phi &= \psi'\end{aligned}$$

(“Reduction constraint” $C = \psi' - \phi$; it appears when spatial order is reduced).

How does the wave equation look now? Hint: Write as system with $\dot{A}$ in LHS.

First order reduction of the wave equation (in 1+1) II

Substitute $\pi = \dot{\psi}$ and $\phi = \psi'$ into $\ddot{\psi} = \psi''$; apply ∂_t to ϕ definition. The wave equation is now expressed as a coupled system of 3 first order equations:

$$\begin{aligned} \dot{\psi} &= \pi \quad \rightarrow \text{This is just an ODE. Let's focus on the others.} \\ \left. \begin{aligned} \dot{\phi} &= \pi' \\ \dot{\pi} &= \phi' \end{aligned} \right\} \rightarrow \text{Coupled system of PDEs. Consider following variable change:} \end{aligned}$$

$$\begin{aligned} l &= \phi + \pi \\ r &= \phi - \pi \end{aligned}$$

The two PDEs transform into:

$$\begin{aligned} \dot{l} &= l' & (\equiv \partial_t l = \partial_x l) \\ \dot{r} &= -r' & (\equiv \partial_t r = -\partial_x r) \end{aligned}$$

l and r are characteristic variables.

They allow to express the system as decoupled scalar equations.

Characteristic decomposition

Equations decouple thanks to careful choice of variables (characteristic ones):

$$\left. \begin{aligned} \dot{\phi} &= \pi' \\ \dot{\pi} &= \phi' \end{aligned} \right\} \rightarrow \left. \begin{aligned} \dot{l} &= l' \\ \dot{r} &= -r' \end{aligned} \right\}$$

$$\partial_t \mathbf{U} = A[\partial_x] \mathbf{U} \quad \leftrightarrow \quad \partial_t \mathbf{V} = D[\partial_x] \mathbf{V} \quad \text{with } D \text{ diagonal:}$$

$$\mathbf{U} = \begin{pmatrix} \phi \\ \pi \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \partial_x, \quad \mathbf{V} = \begin{pmatrix} \phi + \pi \\ \phi - \pi \end{pmatrix} = \begin{pmatrix} l \\ r \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \partial_x$$

Some generalization of this trick works for all “good” hyperbolic PDEs. Hyperbolic PDEs are (intuitively) those which have a well-posed initial value problem with finite speed of propagation. In our case here:

- Well-posedness: There exists a unique solution that depends continuously on the given data (and it cannot grow unboundedly). The equations at hand (advection equations) satisfy this condition. See next slide.
- Finite speed: Solutions to the advection equation are just travelling waves. More on second to next slide ...

Mode solutions to the advection equation

Consider the advection equation $\dot{l} = l'$ with initial data

$$l(0, x) = f(x) = e^{i\omega x}.$$

Its solution is:

$$l(t, x) = e^{i\omega(t+x)}$$

Observe:

$$|l| = |f| = 1$$

It thus satisfies that

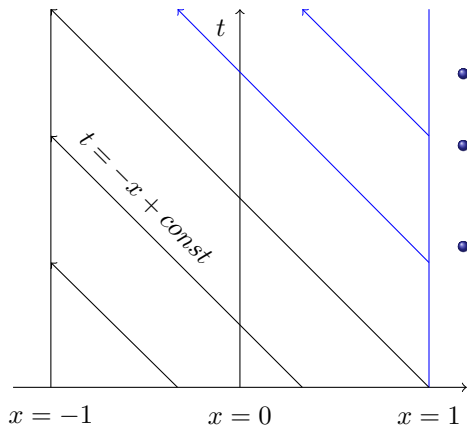
$$|l| \leq K e^{\alpha t} |f|$$

for some K, α independent of f (ω is the only freedom in f).

This means continuous dependence on given data. This can be thought as an energy conservation statement or, more generally, as a certain restriction on the rate of growth of solutions. In particular, there must be a fastest rate of growth for all initial data (K, α independent of ω , in this case).

Left-bound advection equation

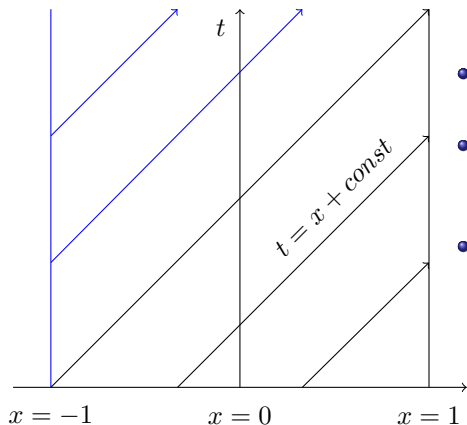
Solution to $\partial_t l = \partial_x l \equiv \dot{l} = l'$ is a wave $l = f(t + x)$ propagating to the left with unity speed. It is constant on lines $t + x = \text{const}$.



- Initial data given on $x \in [-1, 1]$.
- In interior and left boundary (on $x \in [-1, 1)$) evolve with $\partial_t l = \partial_x l$.
- On $x = 1$ specify boundary condition:
 - ▶ Periodic: $l(x = 1) = l(x = -1)$.
 - ▶ Open: $l(x = 1) = 0$.

Right-bound advection equation

Solution to $\partial_t r = -\partial_x r \equiv \dot{r} = -r'$ is a wave $r = f(t - x)$ propagating to the right with unity speed. It is constant on lines $t - x = \text{const.}$

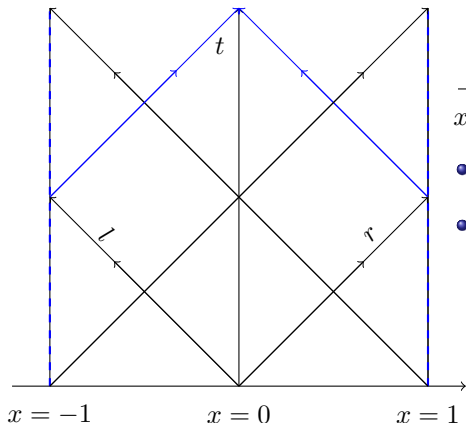


- Initial data given on $x \in [-1, 1]$.
- In interior and right boundary (on $x \in (-1, 1]$) evolve with $\partial_t r = -\partial_x r$.
- On $x = -1$ set boundary condition:
 - ▶ Periodic: $r(x = -1) = r(x = 1)$.
 - ▶ Open: $r(x = -1) = 0$.

Wave equation: two advection equations

The behaviour of the wave equation is the superposition of both fields l and r .

Propagation near the boundaries:



- Initial data given on $x \in [-1, 1]$.

- Boundary conditions:

- ▶ Open:

$$\begin{aligned} r(t, x = -1) &= 0 \\ l(t, x = 1) &= 0 \end{aligned}$$

- ▶ Reflective/Absorbing:

$$\begin{aligned} r(t, x = -1) &= k_0 l(t, x = -1) \\ l(t, x = 1) &= k_1 r(t, x = 1) \end{aligned}$$

Boundary conditions

Apply ∂_t to $r(t, x = -1) = k_0 l(t, x = -1)$ and $l(t, x = 1) = k_1 r(t, x = 1)$:

$$\begin{aligned}\dot{r}(t, x = -1) &= k_0 \dot{l}(t, x = -1) = k_0 l'(t, x = -1) \\ \dot{l}(t, x = 1) &= k_1 \dot{r}(t, x = 1) = -k_1 r'(t, x = 1)\end{aligned}$$

Undo the change from $r = \phi - \pi$ and $l = \phi + \pi$ back to the original ϕ and π :

$$\begin{aligned}\phi &= \frac{1}{2}(l + r) & \rightarrow & \quad \dot{\phi} = \frac{1}{2}(\dot{l} + \dot{r}) \\ \pi &= \frac{1}{2}(l - r) & \rightarrow & \quad \dot{\pi} = \frac{1}{2}(\dot{l} - \dot{r})\end{aligned}$$

- At $x = -1$:

$$\begin{aligned}\dot{\phi} &= \frac{1}{2}\dot{l} + \frac{1}{2}\dot{r} = \frac{1}{2}l' + \frac{k_0}{2}l' = \frac{1+k_0}{2}l' = \frac{1+k_0}{2}(\phi' + \pi') \\ \dot{\pi} &= \frac{1}{2}\dot{l} - \frac{1}{2}\dot{r} = \frac{1}{2}l' - \frac{k_0}{2}l' = \frac{1-k_0}{2}l' = \frac{1-k_0}{2}(\phi' + \pi')\end{aligned}$$

- At $x = 1$:

$$\begin{aligned}\dot{\phi} &= \frac{1}{2}\dot{l} + \frac{1}{2}\dot{r} = -\frac{k_1}{2}r' - \frac{1}{2}r' = -\frac{1+k_1}{2}r' = -\frac{1+k_1}{2}(\phi' - \pi') \\ \dot{\pi} &= \frac{1}{2}\dot{l} - \frac{1}{2}\dot{r} = -\frac{k_1}{2}r' + \frac{1}{2}r' = \frac{1-k_1}{2}r' = \frac{1-k_1}{2}(\phi' - \pi')\end{aligned}$$

Numerical approximation using difference methods

(Following chapter 3 of Kreiss and Busenhardt).

Consider the advection equation

$$u_t = u' \quad \text{for} \quad -\infty < x < \infty \quad \text{and} \quad t \geq 0$$

with initial data $u(0, x) = f(x)$. Assume that $f(x)$, and thus also $u(t, x)$, is 2π -periodic.

Discretize in space but keep time continuous (“semi-discrete system”).

Write the grid interval as

$$h = \frac{2\pi}{2N+1}, \quad N \in \mathbb{N} \quad \left(h = \frac{2\pi}{N+1} \right)$$

which defines a uniform grid

$$x_\nu = \nu h, \quad \nu = 0, \pm 1, \pm 2, \dots \pm N \quad (\nu = 0, 1, 2, \dots N)$$

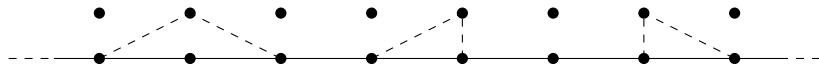
A discrete function v_ν defined on the grid is called a gridfunction.

Discretized finite difference operators

Define a translation operator for $p = 0, \pm 1, \pm 2, \dots$ as $E^p v_\nu = v_{\nu+p}$.

Difference operators:

- Central: $D_0 = \frac{1}{2h}(E - E^{-1})$
- Forward: $D_+ = \frac{1}{h}(E - I)$
- Backward: $D_- = \frac{1}{h}(I - E^{-1})$



$$D_0 v_\nu = \frac{v_{\nu+1} - v_{\nu-1}}{2h}, \quad D_- v_\nu = \frac{v_\nu - v_{\nu-1}}{h}, \quad D_+ v_\nu = \frac{v_{\nu+1} - v_\nu}{h}$$

Simplest approximations to the given advection equation:

$$\left. \begin{aligned} \dot{v}_\nu &= D_- v_\nu \\ \dot{v}_\nu &= D_+ v_\nu \end{aligned} \right\} \rightarrow \text{First order accurate in } h.$$

$$\dot{v}_\nu = D_0 v_\nu \rightarrow \text{Second order accurate. Higher order approx. are also possible.}$$

Convergence order of difference operators

In general, any given finite difference approximation will have an error:

$$Dv_\nu = v'(x_\nu) + O(h^n) \approx v'(x_\nu) + Ch^n + \dots$$

(Equivalent expression for higher derivatives: $D^m v_\nu \approx v^{(m)}(x_\nu) + Ch^n + \dots$).

For the difference operators we are considering here:

$$D_- v_\nu = v'(x_\nu) + O(h), \quad D_+ v_\nu = v'(x_\nu) + O(h), \quad D_0 v_\nu = v'(x_\nu) + O(h^2)$$

Do these approximations work? What does that even mean?

Solutions to the approximation should converge to those of the continuum system as $h \rightarrow 0$.

$$v(x_\nu) \approx v_\nu + Ch^n \quad \text{with} \quad n > 0$$

Convergence of difference operators I

Look for simple wave solutions of the form

$$v_\nu(t) = e^{i\omega x_\nu} \hat{v}(t)$$

(This is effectively the discrete Fourier transformation in space).

$$\begin{aligned} D_- e^{i\omega x_\nu} &= \frac{e^{i\omega x_\nu}}{h} - \frac{e^{i\omega(x_\nu - h)}}{h} = e^{i\omega x_\nu} \left(\frac{1 - e^{-i\omega h}}{h} \right) \\ &= e^{i\omega x_\nu} \left(\frac{1 - \cos \omega h}{h} + \frac{i \sin \omega h}{h} \right) \end{aligned}$$

$$D_+ e^{i\omega x_\nu} = e^{i\omega x_\nu} \left(\frac{e^{i\omega h} - 1}{h} \right) = e^{i\omega x_\nu} \left(\frac{\cos \omega h - 1}{h} + \frac{i \sin \omega h}{h} \right)$$

$$D_0 e^{i\omega x_\nu} = e^{i\omega x_\nu} \left(\frac{e^{i\omega h} - e^{-i\omega h}}{h} \right) = e^{i\omega x_\nu} \frac{i \sin \omega h}{h}$$

Convergence of difference operators II

Putting the equations together:

$$\partial_t \hat{v} = \left(\frac{1 - \cos \omega h}{h} + \frac{i \sin \omega h}{h} \right) \hat{v} \quad (D_-)$$

$$\partial_t \hat{v} = \left(\frac{\cos \omega h - 1}{h} + \frac{i \sin \omega h}{h} \right) \hat{v} \quad (D_+)$$

$$\partial_t \hat{v} = \frac{i \sin \omega h}{h} \hat{v} \quad (D_0)$$

(Solutions are going to be of the form $\hat{v} = e^{\lambda t} \hat{v}_0$ with $\partial_t \hat{v} = \lambda \hat{v}$).

Consider $\omega = \frac{\pi}{2h}$, then $\frac{1 - \cos \omega h}{h} = \frac{1}{h} \rightarrow$ there are simple wave solutions that grow arbitrarily fast as $h \rightarrow 0$. Unstable! This is a bad discretization - why? (Upwinding vs. downwinding).

The solutions to the other approximations do not grow, thus those approximations are called stable.

Stability $\leftrightarrow$ approximation as well-posedness $\leftrightarrow$ continuum problem.

Method of Lines

Take a PDE with a generic form

$$\partial_t q + A^i(q) \partial_i B(q) = S(q)$$

and write it as

$$\partial_t q = L(q) \quad \rightarrow \quad \text{ODE}$$

This works if the spatial derivatives are discretized.

Example: advection equation

$$\partial_t u = \partial_x u$$

Discretize in space:

$$\begin{aligned} \partial_t v_\nu &= D_0 v_\nu \\ \partial_t v_\nu &= \frac{1}{2h} (v_{\nu+1} - v_{\nu-1}) \end{aligned}$$

Integrate in time as a system of ODEs.

Euler method vs Runge-Kutta 4

Consider a differential equation of the form $\partial_t u = \text{rhs}(u, t)$ and time-step Δt .

- A very simple choice is the Euler method (1st order of convergence):

$$u^{n+1} = u^n + \Delta t \text{ rhs}(u^n, t_n) .$$

- Generally it is more convenient to use more elaborate methods of higher order, such as the Runge-Kutta (RK) methods (with intermediate time-steps). The most common one is the 4th order RK with 3 time-levels:

$$\begin{aligned} k_1 &= \Delta t \text{ rhs}(u^n, t_n), \\ k_2 &= \Delta t \text{ rhs}(u^n + k_1/2, t_n + \Delta t/2), \\ k_3 &= \Delta t \text{ rhs}(u^n + k_2/2, t_n + \Delta t/2), \\ k_4 &= \Delta t \text{ rhs}(u^n + k_3, t_n + \Delta t), \\ u^{n+1} &= u^n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

This method is 4th order of convergence.

Courant-Friedrichs-Lewy (CFL) condition I

The Courant number C is the quotient of the time-step Δt and the time interval given by $\frac{\Delta x}{v}$, where v is the maximum propagation speed of the system.

In the case of explicit time integration algorithms, there is a maximum allowed value of the Courant number that will provide a correct and stable solution. This is the Courant-Friedrichs-Lewy (CFL) condition:

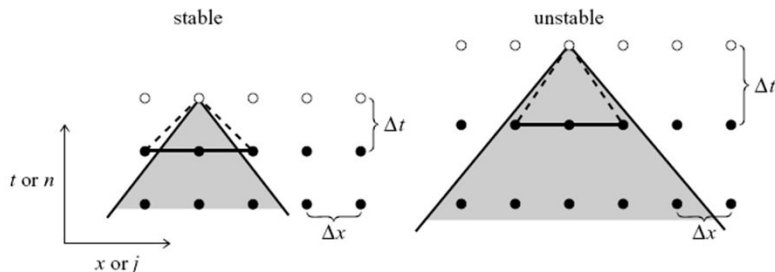
$$C = \frac{v \Delta t}{\Delta x} \leq C_{max}. \quad (1)$$

The value of C_{max} will depend on the exact integration method chosen.

The interpretation of this condition is that the numerical domain of dependence must include the analytical one, so that the information required to integrate to the next time-step is complete.

Courant-Friedrichs-Lewy (CFL) condition II

CFL-condition



Value at a certain point depends on information within some area (shaded) as defined by the PDE. (say advection speed v , wave velocity or speed of light.) These physical points of dependency must be inside the computational used grid points for a stable method.

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Checking convergence of the numerical solution I

Consider a method of convergence order p used with a grid of spacing h . The numerical solution is given by

$$u_h = u_{exact} + \varepsilon h^p$$

where u_{exact} is the exact solution and ε is the h -independent scaled error. Increased by a factor f , the new solution will be $u_{fh} = u_{exact} + \varepsilon (fh)^p$. Increasing the resolution again by f gives $u_{f^2h} = u_{exact} + \varepsilon (f^2h)^p$. Determine the quotient of the differences:

$$\begin{aligned} \frac{u_{f^2h} - u_{fh}}{u_{fh} - u_h} &= \frac{(u_{exact} + \varepsilon f^{2p}h^p) - (u_{exact} + \varepsilon f^p h^p)}{(u_{exact} + \varepsilon f^p h^p) - (u_{exact} + \varepsilon h^p)} = \frac{\varepsilon h^p (f^{2p} - f^p)}{\varepsilon h^p (f^p - 1)} \\ &= \frac{f^p (f^p - 1)}{f^p - 1} = f^p. \end{aligned}$$

If satisfied, then the numerical implementation of the equations converges appropriately and we know that the solution obtained is the correct one.

Checking convergence of the numerical solution II

Convergence order as a function of time using the L^2 norm:

$$p = \log_f \left(\frac{\sqrt{\sum_i (u_{f^2h} - u_{fh})^2}}{\sqrt{\sum_i (u_{fh} - u_h)^2}} \right).$$

Note that the differences are to be taken at exactly the same gridpoints.

For constraints, whose exact value is zero, the expression is even simpler. Denoting the value of the constraint by u , we have that $u_h = \varepsilon h^p$, $u_{fh} = \varepsilon (fh)^p$ and $u_{f^2h} = \varepsilon (f^2h)^p$, so that

$$\frac{u_{f^2h}}{u_{fh}} = \frac{u_{fh}}{u_h} = f^p.$$

For more info, see e.g.:

<https://www.csc.kth.se/utbildning/kth/kurser/DN2255/ndiff13/ConvRate.pdf>